

## Unit 17: Extreme Values on Closed Intervals — Full Solutions

### Warm-up

1.  $f(x) = x^2 - 4x + 3$  on  $[0, 5]$

$f'(x) = 2x - 4$ . Critical point:  $2x - 4 = 0 \Rightarrow x = 2$  (inside  $[0, 5]$ ).

Evaluate:  $f(0) = 3$ .  $f(2) = 4 - 8 + 3 = -1$ .  $f(5) = 25 - 20 + 3 = 8$ .

Absolute max: 8 at  $x = 5$ . Absolute min:  $-1$  at  $x = 2$ .

2.  $f(x) = x^3 - 3x$  on  $[-2, 2]$

$f'(x) = 3x^2 - 3 = 3(x^2 - 1)$ . Critical points:  $x = 1$ ,  $x = -1$  (both inside  $[-2, 2]$ ).

Evaluate:  $f(-2) = -8 + 6 = -2$ .  $f(-1) = -1 + 3 = 2$ .  $f(1) = 1 - 3 = -2$ .  $f(2) = 8 - 6 = 2$ .

Absolute max: 2, occurring at both  $x = -1$  and  $x = 2$ . Absolute min:  $-2$ , occurring at both  $x = -2$  and  $x = 1$ .

3.  $f(x) = x^2 - 6x + 8$  on  $[-1, 4]$

$f'(x) = 2x - 6$ . Critical point:  $x = 3$  (inside  $[-1, 4]$ ).

Evaluate:  $f(-1) = 1 + 6 + 8 = 15$ .  $f(3) = 9 - 18 + 8 = -1$ .  $f(4) = 16 - 24 + 8 = 0$ .

Absolute max: 15 at  $x = -1$ . Absolute min:  $-1$  at  $x = 3$ .

4.  $f(x) = x^3 - 6x^2 + 9x$  on  $[0, 5]$

$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$ . Critical points:  $x = 1$ ,  $x = 3$ .

Evaluate:  $f(0) = 0$ .  $f(1) = 1 - 6 + 9 = 4$ .  $f(3) = 27 - 54 + 27 = 0$ .  $f(5) = 125 - 150 + 45 = 20$ .

Absolute max: 20 at  $x = 5$ . Absolute min: 0, occurring at both  $x = 0$  and  $x = 3$ .

5.  $f(x) = x^2 + 2x - 3$  on  $[-3, 1]$

$f'(x) = 2x + 2$ . Critical point:  $x = -1$ .

Evaluate:  $f(-3) = 9 - 6 - 3 = 0$ .  $f(-1) = 1 - 2 - 3 = -4$ .  $f(1) = 1 + 2 - 3 = 0$ .

Absolute max: 0, occurring at both  $x = -3$  and  $x = 1$ . Absolute min:  $-4$  at  $x = -1$ .

6.  $f(x) = -x^2 + 4x + 1$  on  $[0, 5]$

$f'(x) = -2x + 4$ . Critical point:  $x = 2$ .

Evaluate:  $f(0) = 1$ .  $f(2) = -4 + 8 + 1 = 5$ .  $f(5) = -25 + 20 + 1 = -4$ .

Absolute max: 5 at  $x = 2$ . Absolute min:  $-4$  at  $x = 5$ .

---

### Standard

7.  $f(x) = x^3 - 3x^2 - 9x + 5$  on  $[-2, 4]$

$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$ . Critical points:  $x = 3$ ,  $x = -1$  (both inside  $[-2, 4]$ ).

Evaluate:  $f(-2) = -8 - 12 + 18 + 5 = 3$ .  $f(-1) = -1 - 3 + 9 + 5 = 10$ .  $f(3) = 27 - 27 - 27 + 5 = -22$ .  $f(4) = 64 - 48 - 36 + 5 = -15$ .

Absolute max: 10 at  $x = -1$ . Absolute min:  $-22$  at  $x = 3$ .

8.  $f(x) = x^{2/3}$  on  $[-1, 8]$

$f'(x) = \frac{2}{3}x^{-1/3}$ , which is undefined at  $x = 0$  (and never equals 0 for any  $x$ , since a nonzero fraction raised to  $-1/3$  can never be 0). So  $x = 0$  is a critical point.

Evaluate:  $f(-1) = (-1)^{2/3} = ((-1)^{1/3})^2 = (-1)^2 = 1$ .  $f(0) = 0$ .  $f(8) = 8^{2/3} = (8^{1/3})^2 = 2^2 = 4$ .

Absolute max: 4 at  $x = 8$ . Absolute min: 0 at  $x = 0$ .

9.  $f(x) = x^4 - 2x^2$  on  $[-2, 2]$

$f'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 4x(x - 1)(x + 1)$ . Critical points:  $x = 0$ ,  $x = 1$ ,  $x = -1$ .

Evaluate:  $f(-2) = 16 - 8 = 8$ .  $f(-1) = 1 - 2 = -1$ .  $f(0) = 0$ .  $f(1) = 1 - 2 = -1$ .  $f(2) = 16 - 8 = 8$ .

Absolute max: 8, occurring at both  $x = -2$  and  $x = 2$ . Absolute min:  $-1$ , occurring at both  $x = -1$  and  $x = 1$ .

10.  $f(x) = x + \frac{1}{x}$  on  $[0.5, 4]$

$f'(x) = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2}$ . Setting the numerator to 0:  $x = \pm 1$ ; only  $x = 1$  lies inside  $[0.5, 4]$ . ( $f'$  is undefined at  $x = 0$ , but 0 isn't in this interval, so that's not a concern here.)

Evaluate:  $f(0.5) = 0.5 + 2 = 2.5$ .  $f(1) = 1 + 1 = 2$ .  $f(4) = 4 + 0.25 = 4.25$ .

Absolute max: 4.25 at  $x = 4$ . Absolute min: 2 at  $x = 1$ .

11.  $f(x) = \sin x + \cos x$  on  $[0, \pi]$

$f'(x) = \cos x - \sin x$ . Setting this to 0:  $\cos x = \sin x \Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}$  (the only solution inside  $[0, \pi]$ ).

Evaluate:  $f(0) = 0 + 1 = 1$ .  $f\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2} \approx 1.414$ .  $f(\pi) = 0 + (-1) = -1$ .

Absolute max:  $\sqrt{2}$  at  $x = \frac{\pi}{4}$ . Absolute min:  $-1$  at  $x = \pi$ .

12.  $f(x) = x^3 - 12x$  on  $[-3, 1]$

$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$ . Critical points:  $x = 2$ ,  $x = -2$ ; only  $x = -2$  lies inside  $[-3, 1]$ .

Evaluate:  $f(-3) = -27 + 36 = 9$ .  $f(-2) = -8 + 24 = 16$ .  $f(1) = 1 - 12 = -11$ .

Absolute max: 16 at  $x = -2$ . Absolute min:  $-11$  at  $x = 1$ .

---

Challenge

13.  $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x + 1$  on  $[0, 3]$

$f'(x) = x^2 - 3x + 2 = (x-1)(x-2)$ . Critical points:  $x = 1, x = 2$ .

Evaluate each candidate, using a common denominator of 6 for clarity:

$f(0) = 1$ .

$f(1) = \frac{1}{3} - \frac{3}{2} + 2 + 1 = \frac{2}{6} - \frac{9}{6} + \frac{12}{6} + \frac{6}{6} = \frac{11}{6} \approx 1.833$ .

$f(2) = \frac{8}{3} - 6 + 4 + 1 = \frac{8}{3} - 1 = \frac{8}{3} - \frac{3}{3} = \frac{5}{3} \approx 1.667$ .

$f(3) = 9 - 13.5 + 6 + 1 = 2.5$ .

Absolute max: 2.5 at  $x = 3$ . Absolute min: 1 at  $x = 0$ . Notice both extremes occur at the endpoints, not at either critical point — a great reminder to always check the endpoints, since the critical points here only produced local wiggles in between.

14.  $f(x) = 3x^4 - 4x^3$  on  $[-1, 2]$

$f'(x) = 12x^3 - 12x^2 = 12x^2(x-1)$ . Critical points:  $x = 0, x = 1$ .

Evaluate:  $f(-1) = 3 + 4 = 7$ .  $f(0) = 0$ .  $f(1) = 3 - 4 = -1$ .  $f(2) = 48 - 32 = 16$ .

Absolute max: 16 at  $x = 2$ . Absolute min:  $-1$  at  $x = 1$ .

15.  $f(x) = x^5 - 5x$  on  $[-2, 2]$

$f'(x) = 5x^4 - 5 = 5(x^4 - 1) = 5(x^2 - 1)(x^2 + 1)$ . Since  $x^2 + 1$  is never 0 for real numbers, the only real critical points come from  $x^2 - 1 = 0$ , giving  $x = 1, x = -1$ .

Evaluate:  $f(-2) = -32 + 10 = -22$ .  $f(-1) = -1 + 5 = 4$ .  $f(1) = 1 - 5 = -4$ .  $f(2) = 32 - 10 = 22$ .

Absolute max: 22 at  $x = 2$ . Absolute min:  $-22$  at  $x = -2$ . Again, the true extremes sit at the endpoints, while the critical points only mark local bumps.

16.  $f(x) = (x-1)^{2/3}$  on  $[-7, 9]$

$f'(x) = \frac{2}{3}(x-1)^{-1/3}$ , which is undefined at  $x = 1$  (and never equals 0). So  $x = 1$  is the only critical point.

Evaluate:  $f(-7) = (-8)^{2/3} = ((-8)^{1/3})^2 = (-2)^2 = 4$ .  $f(1) = 0$ .  $f(9) = 8^{2/3} = (8^{1/3})^2 = 2^2 = 4$ .

Absolute max: 4, occurring at both  $x = -7$  and  $x = 9$ . Absolute min: 0 at  $x = 1$  — the sharp cusp point turns out to be the global minimum here.

17.  $f(x) = 2x^3 - 9x^2 + 12x + 5$  on  $[0, 3]$

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2)$ . Critical points:  $x = 1, x = 2$ .

Evaluate:  $f(0) = 5$ .  $f(1) = 2 - 9 + 12 + 5 = 10$ .  $f(2) = 16 - 36 + 24 + 5 = 9$ .  $f(3) = 54 - 81 + 36 + 5 = 14$ .

Absolute max: 14 at  $x = 3$ . Absolute min: 5 at  $x = 0$ . Once again, both true extremes land on the endpoints.

---

Applied

18.  $P(x) = -2x^2 + 40x - 150$  on  $[0, 15]$

$P'(x) = -4x + 40$ . Critical point:  $x = 10$ .

Evaluate:  $P(0) = -150$ .  $P(10) = -200 + 400 - 150 = 50$ .  $P(15) = -450 + 600 - 150 = 0$ .

Absolute max profit: 50 (thousand dollars) at  $x = 10$  units. Absolute min profit:  $-150$  (thousand dollars, a loss) at  $x = 0$  units.

19.  $h(t) = -16t^2 + 96t$  on  $[0, 6]$

$h'(t) = -32t + 96$ . Critical point:  $t = 3$ .

Evaluate:  $h(0) = 0$ .  $h(3) = -144 + 288 = 144$ .  $h(6) = -576 + 576 = 0$ .

Absolute max height: 144 ft at  $t = 3$  seconds. Absolute min height: 0 ft, occurring at both  $t = 0$  and  $t = 6$  (the projectile starts and lands at ground level).

20.  $T(t) = -t^2 + 8t + 10$  on  $[0, 10]$

$T'(t) = -2t + 8$ . Critical point:  $t = 4$ .

Evaluate:  $T(0) = 10$ .  $T(4) = -16 + 32 + 10 = 26$ .  $T(10) = -100 + 80 + 10 = -10$ .

Absolute max temperature:  $26^\circ F$  at  $t = 4$  hours. Absolute min temperature:  $-10^\circ F$  at  $t = 10$  hours.

21.  $C(x) = x^3 - 15x^2 + 63x + 100$  on  $[0, 10]$

$C'(x) = 3x^2 - 30x + 63 = 3(x^2 - 10x + 21) = 3(x - 3)(x - 7)$ . Critical points:  $x = 3$ ,  $x = 7$ .

Evaluate:  $C(0) = 100$ .  $C(3) = 27 - 135 + 189 + 100 = 181$ .  $C(7) = 343 - 735 + 441 + 100 = 149$ .  $C(10) = 1000 - 1500 + 630 + 100 = 230$ .

Absolute max cost: \$230 at  $x = 10$  units. Absolute min cost: \$100 at  $x = 0$  units. Interestingly,  $x = 3$  gives a local maximum (\$181) and  $x = 7$  gives a local minimum (\$149) along the way — but neither is the true global extreme, which once again belong to the endpoints.